PREDICTIVE ANALYTICS

PROJECT REPORT

(Project Semester August–December 2026)

Video Game Sales Analysis and Prediction Using Python and Machine Learning

Submitted by

Himanshu Singh

Registration No. 12411444

Programme and Section: B.Tech (Computer Science & Engineering), Section K2P24NK

Course Code: INT234

Under the Guidance of

Dr. Mrinalini Rana (U.Id: 22138)

Discipline of CSE/IT

Lovely School of Computer Science and Engineering

Lovely Professional University, Phagwara

DECLARATION

I, **Himanshu Singh**, student of **B.Tech (Computer Science & Engineering)** under CSE/IT Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

The project titled **"Video Game Sales Analysis and Prediction Using Python and Machine Learning"** has been carried out using the publicly available *Video Games Sales (1980–2024)* dataset. All external tools, libraries and references used in the work have been duly acknowledged.

Date: 31 / 10 / 2026

Registration No. 12411444

Name of the student: **Himanshu Singh**

Signature: _____

CERTIFICATE

This is to certify that **Himanshu Singh** bearing Registration no. 12411444 has completed INT234 project titled, **"Video Game Sales Analysis and Prediction Using Python and Machine Learning"** under my guidance and supervision. To the best of my knowledge, the present work is the result of his original development, effort and study.

Signature and Name of the Supervisor: Dr. Mrinalini Rana (U.Id: 22138)
Designation of the Supervisor
School of Computer Science and Engineering
Lovely Professional University
Phagwara, Punjab.

Date: 31 / 10 / 2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my faculty coordinator, Dr. Mrinalini Rana, for the continuous guidance, constructive feedback and encouragement provided throughout this project. Their suggestions on structuring the analysis, validating the results and presenting the findings helped me improve the quality of this work.

I am thankful to the Lovely School of Computer Science and Engineering and Lovely Professional University, Phagwara, for providing the academic environment, learning resources and computing facilities required to complete this project.

I also acknowledge the open-source community behind Python, pandas, NumPy, Matplotlib, Seaborn and scikit-learn, whose well-documented libraries made the data cleaning, visualization and machine learning workflow possible, and the creators of the publicly available *Video Games Sales (1980–2024)* dataset for making the data accessible for learning and research.

Finally, I thank my family and friends for their motivation and support during the course of this work.

Himanshu Singh

Registration No. 12411444

TABLE OF CONTENTS

1. Introduction **1**
1.1 Background 1
1.2 Problem Statement 1
1.3 Objectives 1
1.4 Scope of the Project 2
1.5 Tools and Technologies 3
1.6 Methodology Overview 3
1.7 Organisation of the Report 4
2. Source of Dataset **5**
2.1 Origin of the Data 5
2.2 Description of Attributes 6
2.3 Coverage and Characteristics 6
2.4 Regional Definitions 7
2.5 Limitations of the Source 7
2.6 Files Used and Generated 7
3. EDA Process **8**
3.1 Loading and Initial Inspection 8
3.2 Missing Values and Duplicates 9
3.3 Data Cleaning Steps 11
3.4 Feature Engineering 13
3.5 First Descriptive Insights 13
3.6 Data-Quality Observations 15
4. Analysis on Dataset **16**
4.1 Analysis 1: Distribution of Games by Genre 16
4.2 Analysis 2: Recorded Sales by Region 17
4.3 Analysis 3: Average Sales by Genre 19
4.4 Analysis 4: Release Trends by Year and Decade 21
4.5 Analysis 5: Critic Score and Sales 23
4.6 Analysis 6: Average Sales by Platform 25
4.7 Analysis 7: Distribution of Total Sales 27
4.8 Analysis 8: Correlation Between Regional Sales 28
4.9 Analysis 9: Predicting Total Sales with Machine Learning 30
5. Conclusion **34**
5.1 Summary of the Work 34
5.2 Key Findings 34
5.3 Recommendations 35
5.4 Limitations 36
5.5 Project Deliverables and Public Availability 36
5.6 Learning Outcomes 38
6. Future Scope **39**
7. References **40**

LIST OF FIGURES

Figure 1: Overall methodology of the project....3
Figure 2: Output of df.info() and df.head() for the raw dataset....8
Figure 3: Last five records and summary statistics of numeric columns....9
Figure 4: Summary statistics including text columns....9
Figure 5: Missing values before cleaning, duplicate count and shape after cleaning....10
Figure 6: Missing values and summary statistics after cleaning....12
Figure 7: Percentage of missing values per column after cleaning....12
Figure 8: Most common genres and platforms in the cleaned dataset....14
Figure 9: Top 10 games by sales and model evaluation output....15
Figure 10: Top 10 genres by number of games....17
Figure 11: Recorded sales by region....19
Figure 12: Top 10 genres by average sales....21
Figure 13: Number of games released by year....22
Figure 14: Games released by decade....23
Figure 15: Critic score and total sales (symlog scale)....24
Figure 16: Average sales by critic score category....25
Figure 17: Top 10 platforms by average sales....26
Figure 18: Distribution of recorded total sales....28
Figure 19: Correlation between regional sales....30
Figure 20: Comparison of MAE, RMSE and R^2 for the four models....33
Figure 21: GitHub repository of the project (Preditive_Analysis_CA2)....37
Figure 22: Commit activity of the repository (GitHub Insights)....37
Figure 23: LinkedIn post summarising the project....38

LIST OF TABLES

Table 1: Tools and libraries used in the project 3
Table 2: Attributes of the raw dataset (64,016 records) 6
Table 3: Data files used and produced by the project 7
Table 4: Missing values in the raw dataset 10
Table 5: Data cleaning operations applied to the raw dataset 11
Table 6: Engineered features 13
Table 7: Top 10 records by recorded total sales (millions of units) 14
Table 8: Ten most frequent genres 17
Table 9: Recorded sales by region (millions of units) 18
Table 10: Top 10 genres by average total sales (games with recorded sales only) 20
Table 11: Games released by decade 22
Table 12: Sales by critic-score category (games with both values) 24
Table 13: Top 10 platforms by average total sales 26
Table 14: Descriptive statistics of total sales (millions of units, n = 18,922) 28
Table 15: Pearson correlation matrix of regional sales 29
Table 16: Regression models and hyper-parameters 31
Table 17: Model performance on the test set (sorted by RMSE) 32

1. Introduction

1.1 Background

The global video game industry has grown from a niche pastime in the early 1980s into one of the largest entertainment sectors in the world. Hardware generations from the Atari 2600 and Nintendo Entertainment System (NES) to the PlayStation 5 and Nintendo Switch have repeatedly changed how games are built, distributed and consumed. For publishers, developers and analysts, understanding which genres, platforms and release strategies are associated with commercial success is a data-driven question that can influence budgeting, marketing and release planning.

Data science provides a systematic way of answering such questions. By combining data cleaning, exploratory data analysis (EDA), visualization and predictive modelling, large collections of historical sales records can be converted into evidence about market behaviour. This project applies that workflow to a dataset of **64,016 video game records released between 1980 and 2024**, using the Python data-science ecosystem.

1.2 Problem Statement

Public video-game sales data is large but incomplete and noisy. Sales figures are missing for most titles, critic scores are available for only a small fraction of games, regional columns are sparsely populated and several categorical fields are inconsistent. The problem addressed in this project is two-fold:

- to **clean, validate and explore** the raw dataset so that reliable insights about genres, platforms, regions, release trends and critic reception can be drawn; and
- to **predict the total sales (in millions of units)** of a game from a small set of descriptive features—critic score, release year, platform and genre—and to compare several regression algorithms on a held-out test set.

1.3 Objectives

1. To load, inspect and profile the raw *Video Games Sales (1980–2024)* dataset and quantify its data-quality issues.
2. To clean the dataset: standardise text, parse dates, validate numeric ranges and handle missing values and duplicates in a transparent, documented way.

3. To engineer additional features such as release year, release decade, critic-score category, regional sales sum and North American sales share.
4. To perform exploratory analysis and produce ten publication-quality visualizations covering genres, regions, platforms, release trends, critic score and sales distribution.
5. To build a reusable scikit-learn preprocessing pipeline and compare K-Nearest Neighbours (KNN), Linear Regression, Decision Tree and Random Forest regressors using MAE, RMSE and R^2.
6. To discuss the limitations imposed by missing data and to suggest improvements for future work.

Within these objectives, five dataset-specific questions are investigated in Chapter 4:

- **Question 1 (Analysis 1):** Which genres have the largest number of games?
- **Question 2 (Analysis 2):** Which region contributes most to recorded sales?
- **Question 3 (Analyses 3 and 6):** Which genres and platforms achieve the highest average sales per game?
- **Question 4 (Analysis 4):** How has the number of game releases changed over the years and decades?
- **Question 5 (Analysis 5):** Are higher critic scores associated with higher sales?

In addition, Analysis 7 studies the distribution of total sales, Analysis 8 the correlation between regional markets and Analysis 9 the prediction of total sales with four machine-learning models.

1.4 Scope of the Project

The project covers the complete analytical life cycle on a single tabular dataset: data acquisition, cleaning, feature engineering, EDA, visualization, supervised regression modelling and interpretation. It is an academic project: models are trained on a single random train/test split with fixed hyper-parameters, and the results are intended to demonstrate methodology and to give a realistic baseline rather than a production forecasting system. Deployment, text mining of titles and time-series forecasting are outside the present scope and are listed under future scope (Chapter 6).

1.5 Tools and Technologies

The entire work is implemented in Python. Table 1 summarises the tools used and their role in the project.

Table 1: Tools and libraries used in the project

Tool / Library	Purpose in this project
Python 3	Programming language for the complete workflow
pandas	Data loading, cleaning, date parsing, grouping, aggregation and CSV export
NumPy	Numerical operations, handling of missing values (NaN) and square root for RMSE
Matplotlib	Creation and export of figures (bar, line, scatter and histogram plots)
Seaborn	Statistical plots, colour palettes, histogram with KDE and correlation heat map
scikit-learn	Imputation, scaling, one-hot encoding, pipelines, train/test split, regressors and metrics
Jupyter Notebook / VS Code	Interactive development and execution of the analysis
Git and GitHub	Version control and public hosting of code, data and figures

1.6 Methodology Overview

The work follows the standard data-science pipeline shown in Figure 1. The raw dataset is first profiled and cleaned; new features are derived; exploratory analysis is performed with ten visualizations; and finally the cleaned data is passed through a preprocessing pipeline into four regression models whose performance is compared on unseen data.

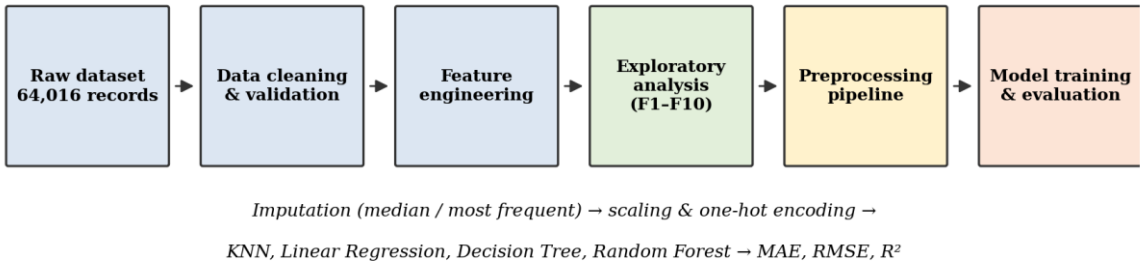


Figure 1: Overall methodology of the project

1.7 Organisation of the Report

Chapter 2 describes the source and structure of the dataset. Chapter 3 documents the EDA and cleaning process. Chapter 4 presents nine individual analyses, each with an introduction, general description, specific requirements/functions/formulas, results and visualization. Chapter 5 concludes the work, Chapter 6 outlines future scope and Chapter 7 lists the references.

2. Source of Dataset

2.1 Origin of the Data

The data used in this project is the publicly available **Video Games Sales (1980–2024)** dataset, obtained as a CSV file (*Video Games Sales (1980-2024) - Raw.csv*). The structure of the file—box-art image paths of the form /games/boxart/…, the PAL region and a *last_update* field—indicates that it is a compiled snapshot of online video-game sales tracking data in which sales are expressed in millions of units. The dataset was obtained from Kaggle, where it is available as an input file of the public notebook *Video Games Sales EDA* (https://www.kaggle.com/code/bhushandivekar/video-games-sales-eda). The sales figures follow the structure of VGChartz-based Kaggle video-game sales data (for example the Kaggle dataset *Video Game Sales 2024* by asaniczka). In this project the file is stored locally as *Video Games Sales (1980-2024) - Raw.csv* and was downloaded on 28 / 10 / 2026 (see Reference [12]).

The raw file contains **64,016 rows and 14 columns**, with about 6.8 MB of memory usage once loaded into pandas. Each row represents one game released on one platform; the same title can therefore appear several times (for example, *Grand Theft Auto V* appears separately for PS3, PS4 and Xbox 360).

2.2 Description of Attributes

Table 2 lists every attribute of the raw dataset with its data type and the number of non-null values reported by *df.info()* (Figure 2 in Chapter 3).

Table 2: Attributes of the raw dataset (64,016 records)

Column	Type	Non-null	Description
img	text	64,016	Relative path of the game's box-art image
title	text	64,016	Name of the game
console	text	64,016	Platform on which the game was released (81 distinct values)
genre	text	64,016	Genre of the game (20 distinct values)
publisher	text	64,016	Publishing company
developer	text	63,999	Development studio (17 missing)
critic_score	float	6,678	Average critic review score on a 1–10 scale
total_sales	float	18,922	Worldwide sales in millions of units
na_sales	float	12,637	Sales in North America (millions)
jp_sales	float	6,726	Sales in Japan (millions)
pal_sales	float	12,824	Sales in PAL regions – Europe, Africa, Australia (millions)
other_sales	float	15,128	Sales in the rest of the world (millions)
release_date	text	56,965	Release date in DD-MM-YYYY format
last_update	text	17,879	Date on which the record was last updated

2.3 Coverage and Characteristics

- **Time span:** release dates in the cleaned data range from 1980 to 2024, with a median release year of 2008 and a mean of about 2006. The raw file also contains 67 records dated before 1980 (the earliest in 1971); these were set to missing during cleaning (Section 3.3).
- **Diversity:** 39,798 unique titles, 81 platforms, 20 genres and 3,383 distinct publisher values (including the placeholder "Unknown").
- **Most frequent platforms:** PC (12,617 records), PS2 (3,565), DS (3,288), PS4 (2,878) and PS (2,707).
- **Sparse numeric fields:** only 29.6% of the records have a recorded total sales value and only 10.4% have a critic score.

2.4 Regional Definitions

Sales are split into four regional columns. *na_sales* covers North America (United States and Canada), *jp_sales* covers Japan, *pal_sales* covers the PAL territories (mainly Europe and Australia) and *other_sales* covers all remaining markets. For a record where all four regional values are present, the total should equal their sum, a relationship that is used later in the cleaning step and in the correlation analysis.

2.5 Limitations of the Source

- Sales figures are estimates compiled by VGChartz from public tracking, not audited publisher accounts, and digital-only sales are often under-represented.
- Sales and critic scores are missing for the majority of records, so any analysis of sales is restricted to a minority subset of games that are likely to be better known and more commercially successful.
- Regional columns are filled for different subsets of games (for example only 6,726 records have Japan sales), which makes totals across regions not strictly comparable.
- The *genre* field contains broad catch-all categories such as “Misc”, and 8,842 records have an unknown publisher.

2.6 Files Used and Generated

Besides the raw input, the project script generates cleaned and feature-enriched versions of the data, ten figures and a model-evaluation table, as listed in Table 3.

Table 3: Data files used and produced by the project

File	Description
Video Games Sales (1980-2024) - Raw.csv	Original dataset (64,016 × 14)
Video_Games_Sales_Cleaned.csv	Cleaned dataset after text, date and numeric validation
Video_Games_Sales_Featured.csv	Cleaned dataset with five engineered features (64,016 × 19)
Model_Evaluation_Results.csv	MAE, RMSE and R^2 of the four regression models
F1 – F10 (.png)	Ten exploratory visualizations exported at 300 dpi
Project.py / Video_Games_Notebook.ipynb	Complete Python script and notebook of the analysis

3. EDA Process

Exploratory data analysis was carried out in a fixed order: inspect the structure, quantify missing values and duplicates, clean and validate each column type, engineer features and finally summarise the cleaned data. Every step is implemented in *Project.py* and its console output is reproduced below.

3.1 Loading and Initial Inspection

The dataset was read with *pd.read_csv()*. The methods *df.info()*, *df.head()*, *df.tail()*, *df.describe()* and *df.describe(include="all")* were used to understand the structure. Figure 2 shows that the file has 64,016 entries and 14 columns of type float64 (6) and text (8), and displays the first five rows.

```
Dataset information:
<class 'pandas.DataFrame'>
RangeIndex: 64016 entries, 0 to 64015
Data columns (total 14 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   img           64016 non-null  str
 1   title         64016 non-null  str
 2   console       64016 non-null  str
 3   genre         64016 non-null  str
 4   publisher     64016 non-null  str
 5   developer     63999 non-null  str
 6   critic_score  6678 non-null   float64
 7   total_sales   18922 non-null  float64
 8   na_sales      12637 non-null  float64
 9   jp_sales      6726 non-null   float64
 10  pal_sales     12824 non-null  float64
 11  other_sales   15128 non-null  float64
 12  release_date  56965 non-null  str
 13  last_update   17879 non-null  str
dtypes: float64(6), str(8)
memory usage: 6.8 MB

First five rows:
                                            img  ... last_update
0  /games/boxart/full_6510540AmericaFrontccc.jpg  ...         NaN
1  /games/boxart/full_5563178AmericaFrontccc.jpg  ...  03-01-2018
2                   /games/boxart/827563ccc.jpg  ...         NaN
3  /games/boxart/full_9218923AmericaFrontccc.jpg  ...         NaN
4  /games/boxart/full_4990510AmericaFrontccc.jpg  ...  14-01-2018

[5 rows x 14 columns]
```

Figure 2: Output of df.info() and df.head() for the raw dataset

The last five records and the numeric summary are shown in Figure 3. Among the 18,922 games with sales, the mean total sales is 0.349 million units but the maximum is 20.32 million, a first sign of a heavily right-skewed distribution. Critic scores (6,678 values) average 7.22 with a standard deviation of 1.46 and span the full 1–10 range.

```
Last five rows:
                                                  img  ... last_update
64011  /games/boxart/full_2779838AmericaFrontccc.jpg  ...  28-01-2019
64012  /games/boxart/full_8031506AmericaFrontccc.jpg  ...  09-05-2020
64013  /games/boxart/full_6553045AmericaFrontccc.jpg  ...  09-05-2020
64014    /games/boxart/full_6012940JapanFrontccc.png  ...  24-02-2019
64015                      /games/boxart/default.jpg  ...  29-09-2023

[5 rows x 14 columns]

Summary statistics:
       critic_score   total_sales  ...     pal_sales   other_sales
count   6678.000000  18922.000000  ...  12824.000000  15128.000000
mean       7.220440      0.349113  ...      0.149472      0.043041
std        1.457066      0.807462  ...      0.392653      0.126643
min        1.000000      0.000000  ...      0.000000      0.000000
25%        6.400000      0.030000  ...      0.010000      0.000000
50%        7.500000      0.120000  ...      0.040000      0.010000
75%        8.300000      0.340000  ...      0.140000      0.030000
max       10.000000     20.320000  ...      9.850000      3.120000

[8 rows x 6 columns]
```

Figure 3: Last five records and summary statistics of numeric columns

Including text columns in the summary (Figure 4) shows that the most common image path is the placeholder */games/boxart/default.jpg*, used by 7,810 records, and that the *last_update* field contains 1,545 unique dates with 06-01-2018 occurring most often (165 times).

```
Summary statistics:
       critic_score   total_sales  ...     pal_sales   other_sales
count   6678.000000  18922.000000  ...  12824.000000  15128.000000
mean       7.220440      0.349113  ...      0.149472      0.043041
std        1.457066      0.807462  ...      0.392653      0.126643
min        1.000000      0.000000  ...      0.000000      0.000000
25%        6.400000      0.030000  ...      0.010000      0.000000
50%        7.500000      0.120000  ...      0.040000      0.010000
75%        8.300000      0.340000  ...      0.140000      0.030000
max       10.000000     20.320000  ...      9.850000      3.120000

[8 rows x 6 columns]

Summary statistics including text columns:
                              img  ... last_update
count                       64016  ...       17879
unique                      56177  ...        1545
top     /games/boxart/default.jpg  ...  06-01-2018
freq                         7810  ...         165
mean                          NaN  ...         NaN
std                           NaN  ...         NaN
min                           NaN  ...         NaN
25%                           NaN  ...         NaN
50%                           NaN  ...         NaN
75%                           NaN  ...         NaN
max                           NaN  ...         NaN

[11 rows x 14 columns]
```

Figure 4: Summary statistics including text columns

3.2 Missing Values and Duplicates

Counting missing values with *df.isna().sum()* shows that the dataset is dominated by gaps in the numeric columns (Figure 5, Table 4). Text columns are almost complete: only 17 developer values are missing. In contrast, critic scores are missing in 89.57% of the rows and total sales in 70.44%. The regional columns are even sparser, with Japan missing for 89.49% of records. The check *df.duplicated().sum()* returned **0** exact duplicate rows.

```
Missing values before cleaning:
img                 0
title               0
console             0
genre               0
publisher           0
developer          17
critic_score    57338
total_sales     45094
na_sales        51379
jp_sales        57290
pal_sales       51192
other_sales     48888
release_date     7051
last_update     46137
dtype: int64

Exact duplicate rows: 0

Rows removed for missing title, console, or genre: 0

Cleaned dataset shape: (64016, 14)
```

Figure 5: Missing values before cleaning, duplicate count and shape after cleaning

Table 4: Missing values in the raw dataset

Column	Missing	Missing (%)	Column	Missing	Missing (%)
img / title / console / genre / publisher	0	0.00	jp_sales	57,290	89.49
developer	17	0.03	pal_sales	51,192	79.97
critic_score	57,338	89.57	other_sales	48,888	76.37
total_sales	45,094	70.44	release_date	7,051	11.01
na_sales	51,379	80.26	last_update	46,137	72.07

Because most missing values are structural (a score or a sales estimate simply was never recorded for that game), deleting all incomplete rows would discard almost the entire dataset. The cleaning strategy therefore keeps rows wherever possible, marks gaps explicitly, and handles missing values only at the stage where a specific analysis or model needs them.

3.3 Data Cleaning Steps

The cleaning operations, their implementation and the reasoning behind each are summarised in Table 5.

Table 5: Data cleaning operations applied to the raw dataset

Step	Implementation	Reason / outcome
1. Text standardisation	astype("string"), str.strip(); empty strings replaced with missing	Removes stray whitespace and prevents the same category being counted twice
2. Unknown publisher/developer	fillna("Unknown")	Keeps 17 developer gaps (and publisher gaps) as an explicit category instead of NaN; 8,842 records have publisher “Unknown”
3. Date parsing	pd.to_datetime(format="%d-%m-%Y", errors="coerce")	Converts release_date and last_update to datetime; no value failed to parse
4. Date range check	Release years outside 1980–2024 set to NaT	67 invalid release dates were set to missing
5. Numeric coercion	pd.to_numeric(errors="coerce")	Guarantees numeric type for scores and sales
6. Negative sales	Values < 0 set to NaN	No negative sales were found
7. Critic score range	Scores < 1 or > 10 set to NaN	No out-of-range scores were found
8. Total sales recovery	If total is missing and all four regions exist, total = sum of regions	No such row existed, confirming internal consistency
9. Duplicates & key fields	drop_duplicates(); dropna(subset=[title, console, genre])	0 duplicate rows and 0 rows with missing key fields were removed

The cleaned dataset has the same shape as the raw data, (64,016, 14), because no row needed to be removed. The data was already structurally sound; the main cleaning effort was therefore standardising types and documenting gaps rather than deleting records. Figure 6 shows the missing-value counts and summary statistics after cleaning: the developer column now has zero missing values, while the *release_date* column has 7,118 missing values (7,051 original plus the 67 invalid dates). Figure 7 visualises the share of missing values in every column.

```
Missing values after cleaning:
img                 0
title               0
console             0
genre               0
publisher           0
developer           0
critic_score    57338
total_sales     45094
na_sales        51379
jp_sales        57290
pal_sales       51192
other_sales     48888
release_date     7051
last_update     46137
dtype: int64

Summary statistics after cleaning:
       critic_score  ...                 last_update
count   6678.000000  ...                       17879
mean       7.220440  ...  2020-01-11 00:45:49.683986
min        1.000000  ...         2017-11-28 00:00:00
25%        6.400000  ...         2018-08-08 00:00:00
50%        7.500000  ...         2019-04-21 00:00:00
75%        8.300000  ...         2021-03-30 00:00:00
max       10.000000  ...         2024-01-28 00:00:00
std        1.457066  ...                         NaN

[8 rows x 8 columns]
```

Figure 6: Missing values and summary statistics after cleaning

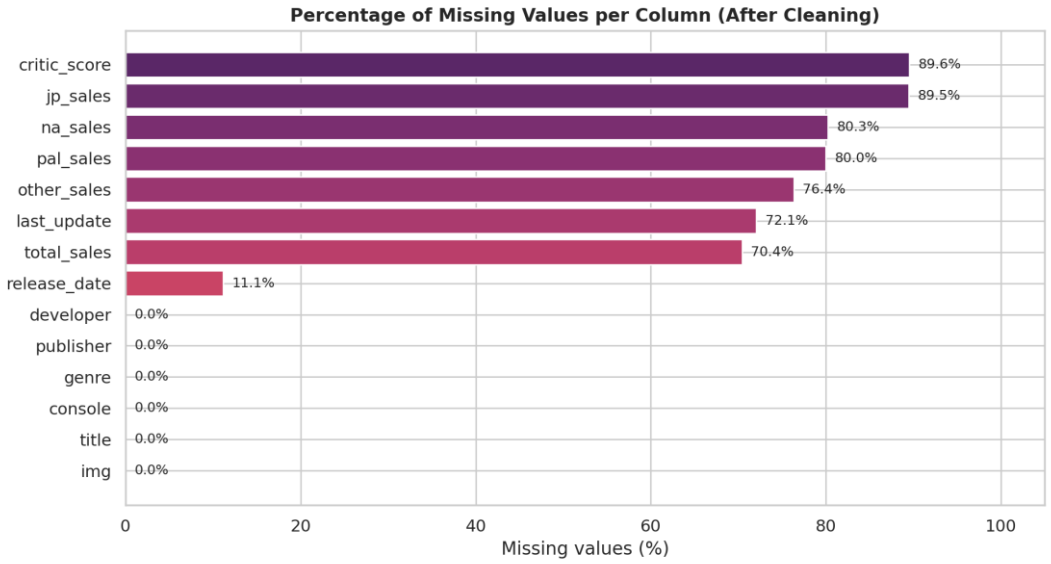


Figure 7: Percentage of missing values per column after cleaning

3.4 Feature Engineering

Five new columns were derived from the cleaned data. They enrich the analysis and are saved in *Video_Games_Sales_Featured.csv*, which therefore has 19 columns.

Table 6: Engineered features

Feature	Derivation	Purpose
release_year	release_date.dt.year	Time-trend analysis and a model input
release_decade	(release_year // 10) × 10	Grouping releases into 1980s … 2020s
critic_score_category	pd.cut(critic_score, [0, 4, 7, 10]) → Low / Average / High	Compare sales across review-quality groups
regional_sales_sum	Row-wise sum of the four regional columns (min_count = 1)	Consistency check of totals
na_sales_share	na_sales / regional_sales_sum	Share of a game's sales that comes from North America (mean ≈ 0.65 where available)

3.5 First Descriptive Insights

Frequency counts of genres and platforms were produced before any plotting (Figure 8). "Misc" (9,304), Action (8,557) and Adventure (6,260) are the most frequent genres, while PC dominates the platform list with 12,617 records, almost four times the next platform (PS2, 3,565).

```
Most common genres:
genre
Misc            9304
Action          8557
Adventure       6260
Role-Playing    5721
Sports          5586
Shooter         5410
Platform        4001
Strategy        3685
Puzzle          3521
Racing          3425
Name: count, dtype: Int64

Most common platforms:
console
PC      12617
PS2      3565
DS       3288
PS4      2878
PS       2707
NS       2337
XBL      2120
PSN      2004
XOne     1963
PS3      1905
Name: count, dtype: Int64
```

Figure 8: Most common genres and platforms in the cleaned dataset

Sorting by total sales shows that the best-selling records are dominated by a few franchises (Table 7, Figure 9). *Grand Theft Auto V* occupies three of the top four places (PS3 with 20.32 million, PS4 with 19.39 million and Xbox 360 with 15.86 million units), and the *Call of Duty* series appears five times in the top ten.

Table 7: Top 10 records by recorded total sales (millions of units)

Rank	Title	Platform	Total sales
1	Grand Theft Auto V	PS3	20.32
2	Grand Theft Auto V	PS4	19.39
3	Grand Theft Auto: Vice City	PS2	16.15
4	Grand Theft Auto V	X360	15.86
5	Call of Duty: Black Ops 3	PS4	15.09
6	Call of Duty: Modern Warfare 3	X360	14.82
7	Call of Duty: Black Ops	X360	14.74
8	Red Dead Redemption 2	PS4	13.94
9	Call of Duty: Black Ops II	X360	13.86
10	Call of Duty: Black Ops II	PS3	13.80

```
Top 10 games by recorded total sales:
                          title console  total_sales
            Grand Theft Auto V     PS3        20.32
            Grand Theft Auto V     PS4        19.39
   Grand Theft Auto: Vice City     PS2        16.15
            Grand Theft Auto V    X360        15.86
     Call of Duty: Black Ops 3     PS4        15.09
Call of Duty: Modern Warfare 3    X360        14.82
       Call of Duty: Black Ops    X360        14.74
         Red Dead Redemption 2     PS4        13.94
    Call of Duty: Black Ops II    X360        13.86
    Call of Duty: Black Ops II     PS3        13.80

Model evaluation:
            Model   MAE  RMSE     R2
              KNN 0.313 0.761  0.209
    Random Forest 0.304 0.771  0.188
Linear Regression 0.364 0.807  0.109
    Decision Tree 0.330 0.928 -0.177

Best-fit model based on lowest RMSE: KNN
MAE: 0.313
RMSE: 0.761
R2: 0.209

Percentage of missing total sales:
70.44 %
```

Figure 9: Top 10 games by sales and model evaluation output

3.6 Data-Quality Observations

- The dataset is clean in terms of structure (no duplicates, valid ranges) but **sparse in content**: 70.44% of total sales are missing.
- Missingness is not random. Games with recorded sales and critic scores tend to be the more prominent releases, so conclusions apply mainly to this better-documented subset.
- Catch-all categories (“Misc”, “Unknown” publisher) limit the interpretability of some groupings.
- Several category averages rest on very few games and must be interpreted with caution; this is highlighted in the relevant analyses in Chapter 4.

4. Analysis on Dataset

This chapter presents nine analyses. Each analysis follows the same five-part structure: (i) introduction, (ii) general description, (iii) specific requirements, functions and formulas, (iv) analysis results and (v) visualization. The first eight analyses are exploratory (Figures 10–19); the ninth is the predictive model comparison (Figure 20). Sales values are in millions of units unless stated otherwise.

4.1 Analysis 1: Distribution of Games by Genre

4.1.1 Introduction

A genre describes the gameplay style of a title, such as Action, Sports or Role-Playing. Counting how many games belong to each genre shows where developers and publishers concentrate their output and which genres are most crowded.

4.1.2 General Description

The analysis counts the number of records per genre in the cleaned dataset and displays the ten most frequent genres as a horizontal bar chart. Only the *genre* column is needed, and it has no missing values, so all 64,016 records are included. The count measures the **supply** of games, not their commercial success; success is examined in later analyses.

4.1.3 Specific Requirements, Functions and Formulas

- **Column required:** genre (text, complete).
- **Functions:** *Series.value_counts()* to count, *head(10)* to select the top ten, *sort_values()* to order bars, *plt.barh()* to draw and *sns.color_palette("viridis")* for colours.

$$\text{Share of genre } g\ (\%) = (n_g / N) \times 100, \quad \text{where } N = 64{,}016$$

```
genre_counts = df["genre"].value_counts().head(10).sort_values()
plt.barh(genre_counts.index, genre_counts.values,
         color=sns.color_palette("viridis", len(genre_counts)))
```

4.1.4 Analysis Results

Table 8 gives the counts and shares. “Misc” is the largest category with 9,304 games (14.5%), followed by Action (8,557; 13.4%) and Adventure (6,260; 9.8%). The top three categories together cover more than one third of the dataset. Role-Playing, Sports and Shooter each have

between 5,400 and 5,700 titles. Because “Misc” is a catch-all label, its leading position partly reflects classification practice rather than a real gameplay genre.

Table 8: Ten most frequent genres

Rank	Genre	Games	Share of dataset
1	Misc	9,304	14.5%
2	Action	8,557	13.4%
3	Adventure	6,260	9.8%
4	Role-Playing	5,721	8.9%
5	Sports	5,586	8.7%
6	Shooter	5,410	8.5%
7	Platform	4,001	6.3%
8	Strategy	3,685	5.8%
9	Puzzle	3,521	5.5%
10	Racing	3,425	5.4%

4.1.5 Visualization

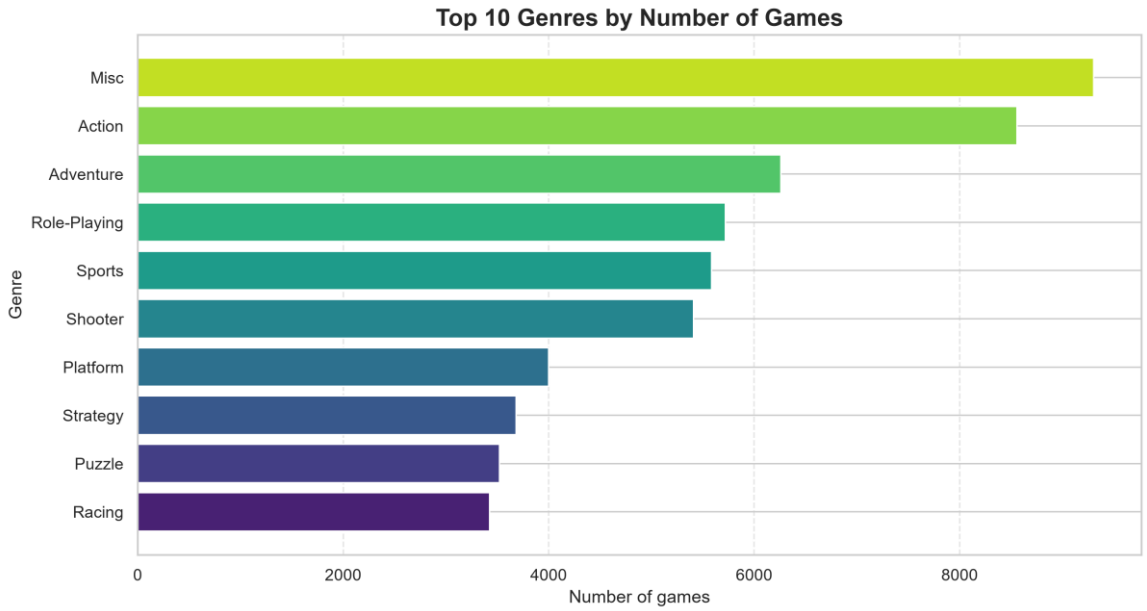


Figure 10: Top 10 genres by number of games

4.2 Analysis 2: Recorded Sales by Region

4.2.1 Introduction

Video game demand differs strongly across the world. Comparing the four sales regions shows which market contributes most to the sales recorded in the dataset.

4.2.2 General Description

The sales values in *na_sales*, *jp_sales*, *pal_sales* and *other_sales* are summed over all games for which a value is recorded, ignoring missing entries. The totals are then sorted and drawn as a column chart. Because the number of games with a recorded value differs between regions (Table 2), the totals describe **recorded** sales, not complete regional markets.

4.2.3 Specific Requirements, Functions and Formulas

- **Columns required:** na_sales, jp_sales, pal_sales, other_sales.
- **Functions:** *df[region_columns].sum()* (NaN values are skipped), *sort_values(ascending=False)* and *plt.bar()*.

$$\text{Regional total } R_r = \Sigma_{i=1..n}\, s_{i,r} \quad \text{(sum over non-missing values only)}$$

$$\text{Regional share (\%)} = R_r \,/\, (R_{NA} + R_{JP} + R_{PAL} + R_{Other}) \times 100$$

```
region_columns = ["na_sales", "jp_sales", "pal_sales", "other_sales"]
regional_sales = df[region_columns].sum().sort_values(ascending=False)
```

4.2.4 Analysis Results

North America is by far the largest region with about 3,345.5 million recorded units, which is 50.7% of the four-region total of 6,601.4 million. PAL territories follow with 1,916.8 million (29.0%). Japan (687.9 million; 10.4%) and the rest of the world (651.1 million; 9.9%) are of similar and much smaller size. Japan's low figure should be read with care: only 6,726 games have a Japan sales value, compared with about 12,600–15,100 for the other regions.

Table 9: Recorded sales by region (millions of units)

Region	Games with value	Total recorded sales	Share
North America (na_sales)	12,637	3,345.52	50.7%
PAL (pal_sales)	12,824	1,916.83	29.0%
Japan (jp_sales)	6,726	687.94	10.4%
Other (other_sales)	15,128	651.12	9.9%
All regions	–	**6,601.41**	**100%**

4.2.5 Visualization

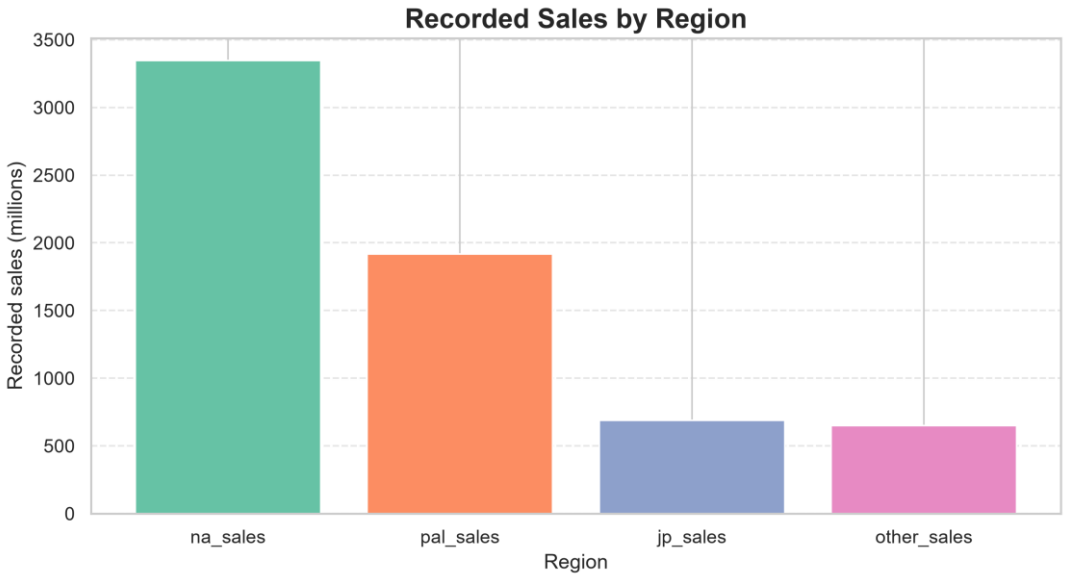


Figure 11: Recorded sales by region

4.3 Analysis 3: Average Sales by Genre

4.3.1 Introduction

Genre frequency (Analysis 1) says how many games exist; average sales say how well a typical game of that genre sells. This analysis identifies the genres whose titles achieve the highest mean sales.

4.3.2 General Description

Rows with missing total sales are removed, the remaining games are grouped by genre, the mean of *total_sales* is calculated for each group and the ten highest means are plotted. The result is sensitive to group size and to a few blockbuster titles, so the number of games behind each mean is reported alongside it.

4.3.3 Specific Requirements, Functions and Formulas

- **Columns required:** genre, total_sales (non-missing).
- **Functions:** *dropna(subset=["total_sales"])*, *groupby("genre")["total_sales"].mean()*, *sort_values()*, *head(10)*.

Mean sales of genre g: $\bar{x}_g = (1 / n_g) \Sigma_{i \in g} total_sales_i$

```
avg_by_genre = (df.dropna(subset=["total_sales"])
                  .groupby("genre")["total_sales"].mean()
                  .sort_values(ascending=False).head(10))
```

4.3.4 Analysis Results

The highest average belongs to **Sandbox** (1.89 million), but this value is based on a single game with recorded sales and therefore says nothing general about the genre. The first reliable result is **Shooter** with 0.670 million units per game across 1,485 titles, followed by Action-Adventure (0.563; 264 titles) and Sports (0.457; 2,597 titles). Action, the second most frequent genre by count, averages only 0.397. In total recorded sales, the leading genres are Sports (1,187.5 million), Action (1,125.9 million) and Shooter (995.5 million). Shooters therefore combine a high average with a large volume, which is consistent with a hit-driven market dominated by franchises such as *Call of Duty*.

Table 10: Top 10 genres by average total sales (games with recorded sales only)

Genre	Average sales (millions)	Games with sales
Sandbox	1.890	1
Shooter	0.670	1,485
Action-Adventure	0.563	264
Sports	0.457	2,597
Action	0.397	2,839
Fighting	0.392	871
Racing	0.367	1,431
Platform	0.367	951
Music	0.352	147
MMO	0.310	30

4.3.5 Visualization

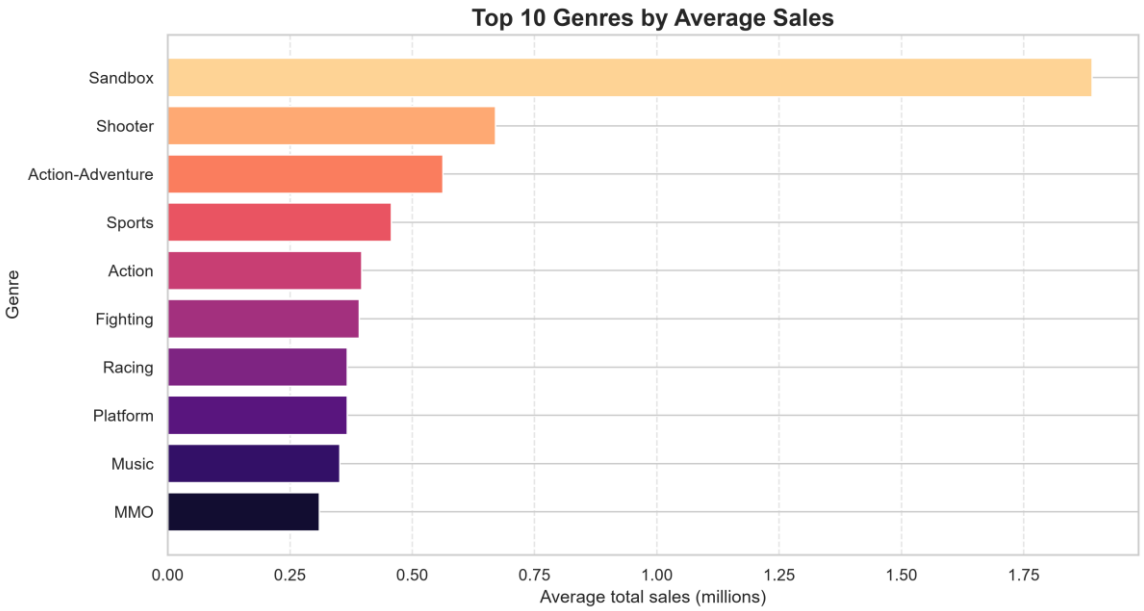


Figure 12: Top 10 genres by average sales

4.4 Analysis 4: Release Trends by Year and Decade

4.4.1 Introduction

Examining the number of games released over time reveals how the industry grew, when platform generations peaked and how complete the dataset is for different periods.

4.4.2 General Description

Release years were derived from the parsed *release_date* column. Games with an unknown or invalid date (7,118 records) are excluded from this analysis, leaving 56,898 games. A line-and-area chart shows releases per year (1980–2024) and a bar chart aggregates them by decade.

4.4.3 Specific Requirements, Functions and Formulas

- **Columns required:** release_date → release_year and release_decade (engineered).
- **Functions:** *dt.year*, integer division for decades, *groupby().size()*, *plt.plot()*, *plt.fill_between()* and *plt.bar()*.

$$release_decade = \lfloor release_year / 10 \rfloor \times 10$$

```
df["release_year"] = df["release_date"].dt.year
df["release_decade"] = (df["release_year"] // 10 * 10).astype("Int64")
games_by_year = (df.dropna(subset=["release_year"])
                   .groupby("release_year").size())
```

4.4.4 Analysis Results

Releases grew slowly through the 1980s, accelerated in the 1990s and peaked in the late 2000s: **2009 is the busiest year with 4,360 games**, and a second smaller peak of roughly 2,900 games appears around 2014. The 2000s (21,812 games) and 2010s (20,610 games) together account for about three quarters of all dated records. After 2020 the counts fall sharply. This fall most likely reflects the way the data was collected—recent releases and digital-only titles are added with a delay and the 2020s decade is incomplete—rather than a real collapse in game production, so the right-hand part of the curve should not be over-interpreted.

Table 11: Games released by decade

Decade	Games	Share of dated records
1980s	1,949	3.4%
1990s	10,355	18.2%
2000s	21,812	38.3%
2010s	20,610	36.2%
2020s (to 2024)	2,172	3.8%
Total	**56,898**	**100%**

4.4.5 Visualization

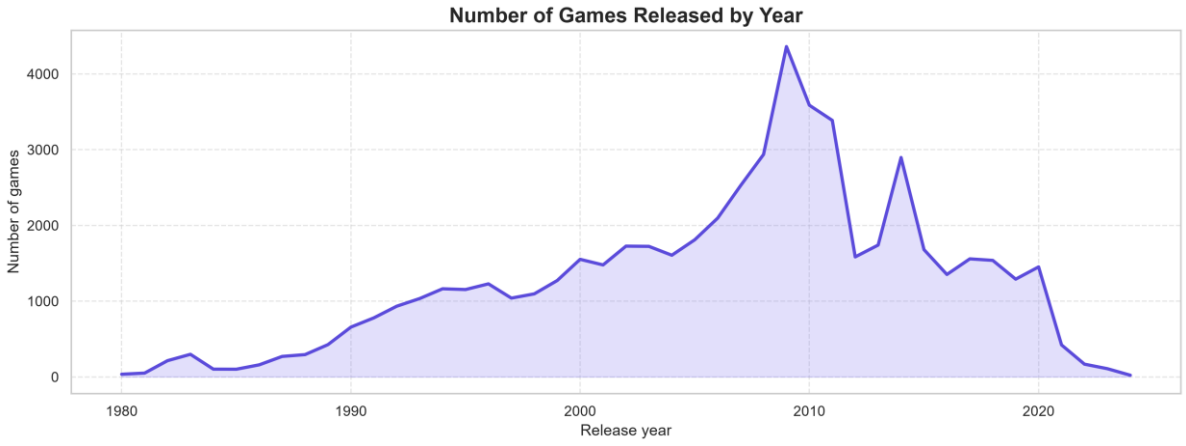


Figure 13: Number of games released by year

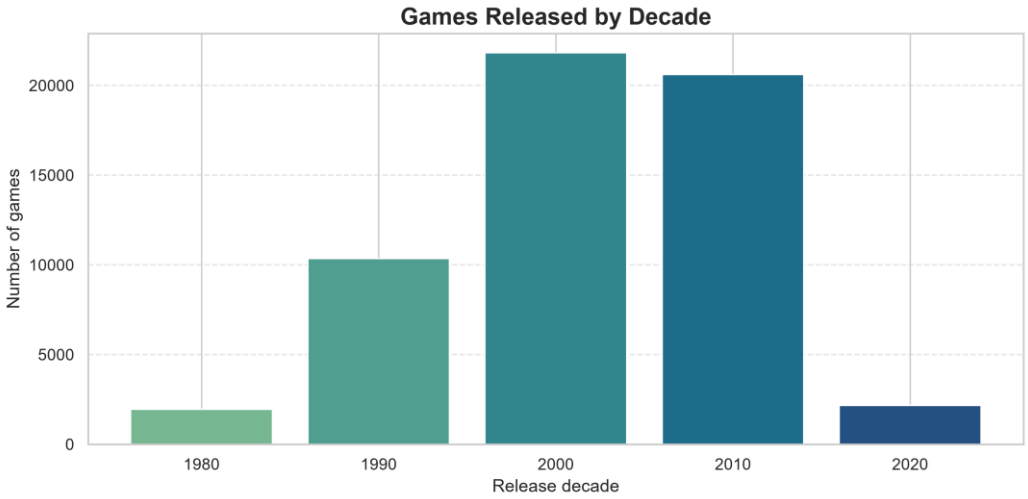


Figure 14: Games released by decade

4.5 Analysis 5: Critic Score and Sales

4.5.1 Introduction

Critic reviews are often assumed to influence purchase decisions. This analysis tests whether higher critic scores are associated with higher sales in the dataset, using both a scatter plot and a categorical comparison.

4.5.2 General Description

Only games with **both** a critic score and total sales can be used. This is a small subset: 4,126 games out of 64,016 (6.4%). Two views are produced. First, a scatter plot of critic score against total sales is drawn with a symmetric-log (symlog) y-axis, because sales range from near zero to over 20 million and a linear axis would compress almost all points at the bottom. Second, critic scores are grouped into three categories—Low (up to 4), Average (above 4 up to 7) and High (above 7)—and the mean sales of each category is compared.

4.5.3 Specific Requirements, Functions and Formulas

- **Columns required:** critic_score, total_sales and the engineered critic_score_category.
- **Functions:** *pd.cut()* with bins [0, 4, 7, 10], *groupby(..., observed=False).mean()*, *plt.scatter()* with a colour map, *plt.yscale("symlog")*.

$$\text{Pearson correlation } r = \Sigma (x_i - \bar{x})(y_i - \bar{y}) / \sqrt{[\, \Sigma (x_i - \bar{x})^2 \cdot \Sigma (y_i - \bar{y})^2 \,]}$$

```
df["critic_score_category"] = pd.cut(
    df["critic_score"], bins=[0, 4, 7, 10],
    labels=["Low", "Average", "High"])
df.groupby("critic_score_category", observed=False)["total_sales"].mean()
```

4.5.4 Analysis Results

The scatter plot (Figure 15) shows a clear but loose upward pattern: the best-selling games almost all have scores above 7, while low-scored games rarely exceed one million units. The Pearson correlation between critic score and total sales on the 4,126 paired records is about **r = 0.28**, a weak-to-moderate positive linear relationship. The category comparison (Figure 16, Table 12) reinforces the pattern: games rated High sell on average **1.00 million** units, about four times the average of Low-rated games (0.24 million) and about 2.4 times that of Average games (0.41 million). The medians show the same order (0.46, 0.21 and 0.12 million) but are much smaller than the means, again indicating a few very large hits. Note that the relationship is an association; popular games may be reviewed more and sold more for reasons unrelated to review quality (for example marketing budget and franchise recognition).

Table 12: Sales by critic-score category (games with both values)

Category	Score range	Games	Mean sales	Median sales
Low	≤ 4	174	0.244	0.12
Average	> 4 to 7	1,625	0.411	0.21
High	> 7 to 10	2,327	1.002	0.46

4.5.5 Visualization

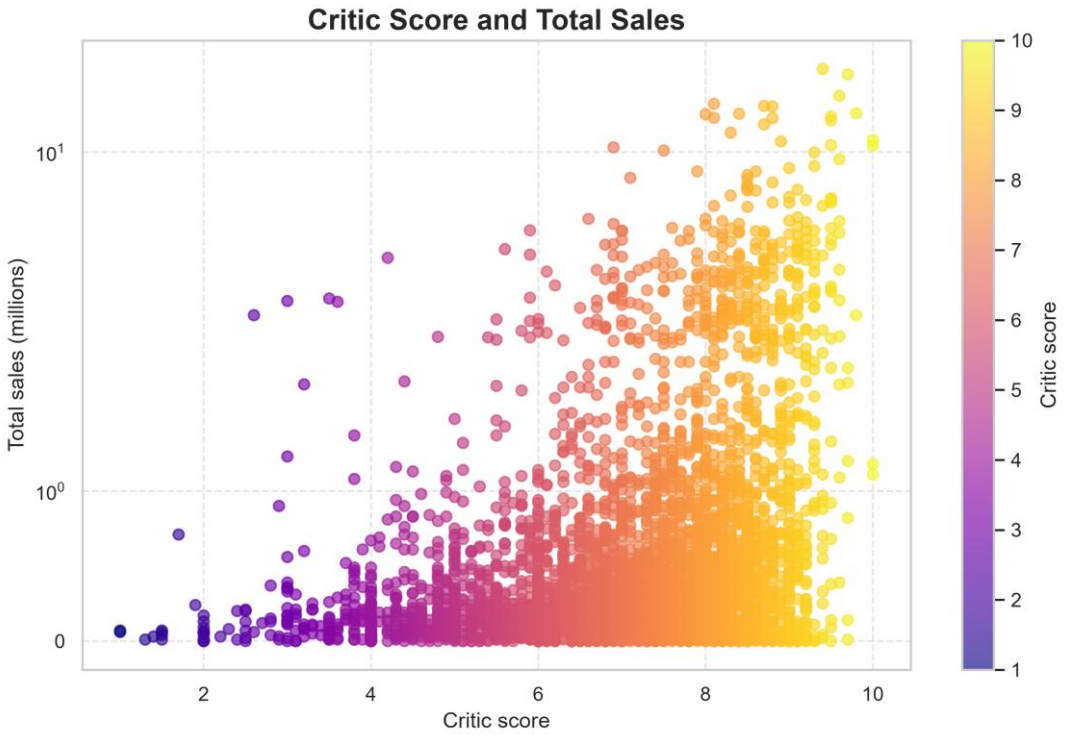


Figure 15: Critic score and total sales (symlog scale)

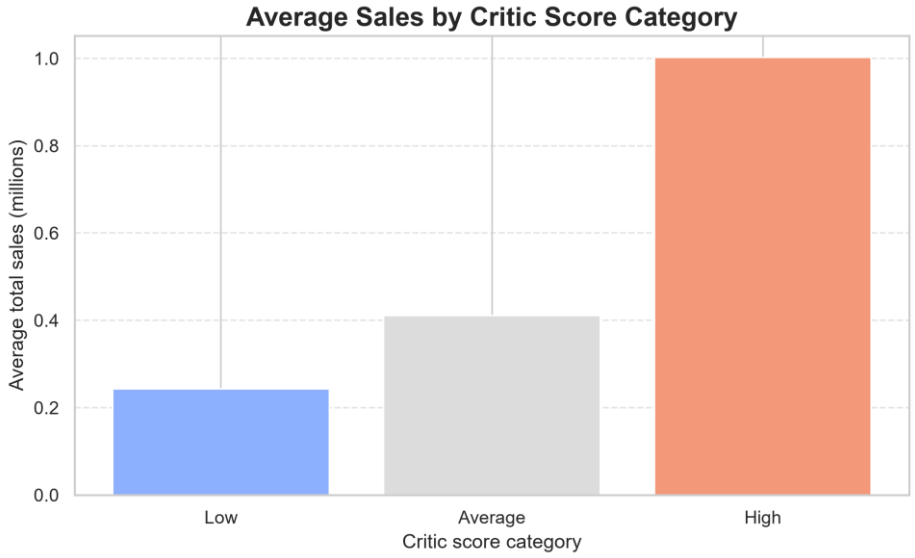


Figure 16: Average sales by critic score category

4.6 Analysis 6: Average Sales by Platform

4.6.1 Introduction

Platforms differ in installed base, software libraries and life span. This analysis asks which platforms produce the highest average sales per game.

4.6.2 General Description

Games with recorded sales are grouped by *console*, the mean total sales per platform is computed and the ten highest averages are displayed. As in Analysis 3, the number of games behind each average determines how reliable it is.

4.6.3 Specific Requirements, Functions and Formulas

- **Columns required:** console, total_sales (non-missing).
- **Functions:** *groupby("console")["total_sales"].mean()*, *sort_values()*, *head(10)*, *plt.barh()* with the "rocket" palette.

$$\textit{Mean sales of platform } p: \quad \bar{x}_p = (1 / n_p)\, \Sigma_{i \in p}\, total_sales_i$$

```
platform_sales = (df.dropna(subset=["total_sales"])
                    .groupby("console")["total_sales"].mean()
                    .sort_values(ascending=False).head(10))
```

4.6.4 Analysis Results

The Game Boy Color (GBC, 1.45 million) and NES (1.00 million) lead the ranking, but they rest on only 3 and 48 games respectively, so they are better described as platforms with a few famous titles than as strong averages. Among platforms with large catalogues the leaders are

Xbox 360 (0.660 million, 1,302 games), **PS3 (0.623 million, 1,348 games)** and **PS4 (0.595 million, 908 games)**. Modern console generations therefore deliver the highest typical sales per title, whereas PS2 (2,126 games) and PS (1,130 games) average about 0.48 million. By cumulative sales the order is PS2 (1,027.8 million), Xbox 360 (859.8 million), PS3 (839.7 million), PS (546.3 million) and PS4 (539.9 million).

Table 13: Top 10 platforms by average total sales

Platform	Average sales (millions)	Games with sales
GBC	1.447	3
NES	0.999	48
GEN (Sega Genesis)	0.697	27
X360	0.660	1,302
PS3	0.623	1,348
2600	0.600	126
PS4	0.595	908
XOne	0.513	524
PS2	0.483	2,126
PS	0.483	1,130

4.6.5 Visualization

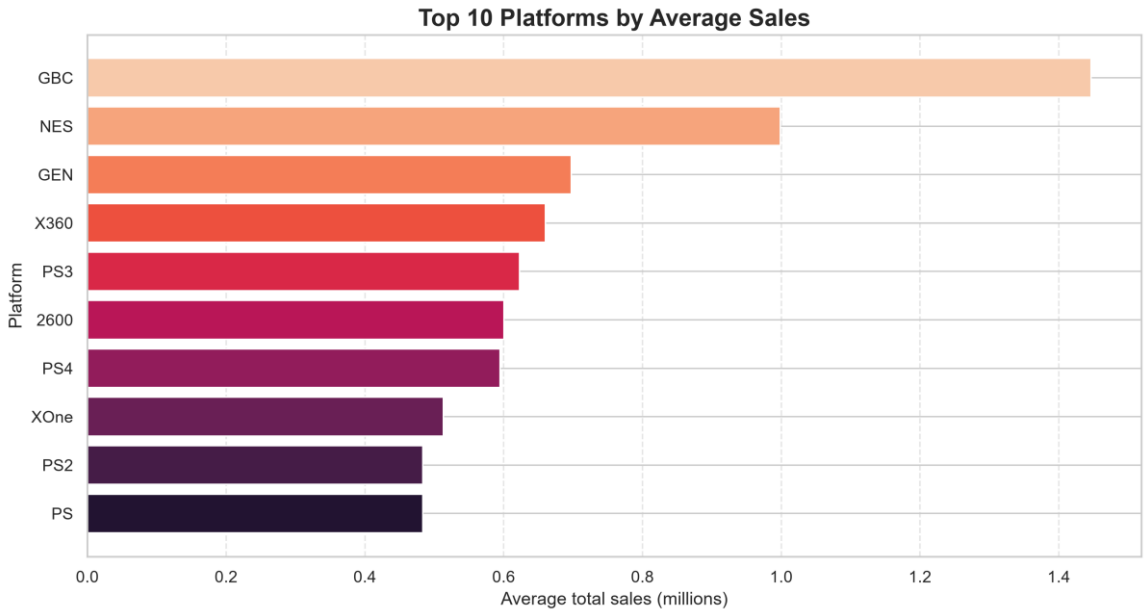


Figure 17: Top 10 platforms by average sales

4.7 Analysis 7: Distribution of Total Sales

4.7.1 Introduction

Before modelling a numeric target it is important to understand its shape. The distribution of total sales shows how concentrated sales are among a few titles and what kind of prediction error can be expected.

4.7.2 General Description

A histogram with 40 bins and a kernel density estimate (KDE) curve is drawn for all 18,922 recorded total-sales values. A symmetric-log x-axis is used so that the long right tail remains visible.

4.7.3 Specific Requirements, Functions and Formulas

- **Column required:** total_sales (non-missing).
- **Functions:** *sns.histplot(bins=40, kde=True)*, *plt.xscale("symlog")*, *Series.describe()* and *quantile()*.

$$Skewness = (1/n)\ \Sigma\ (x_i - \bar{x})^3 / s^3$$

```
sns.histplot(df["total_sales"].dropna(), bins=40, kde=True)
plt.xscale("symlog")
```

4.7.4 Analysis Results

The distribution is extremely right-skewed. The median game sells only 0.12 million units while the mean is 0.349 million, almost three times larger, and the standard deviation (0.807) is more than twice the mean. A quarter of the games sell 0.03 million or less, 90% sell less than 0.83 million, and only 1% exceed 3.54 million. At the top, 1,505 games (8.0%) sold at least one million copies and just 91 games sold five million or more, with a maximum of 20.32 million. About 1,352 games show a recorded value of 0.00, which most likely corresponds to very small sales rounded down to zero in the source. This shape has two consequences for modelling: squared-error metrics such as RMSE will be dominated by the few blockbuster

titles, and any model will find it hard to explain the variance without information that identifies hits (for example franchise, marketing or publisher strength).

Table 14: Descriptive statistics of total sales (millions of units, n = 18,922)

Statistic	Value	Statistic	Value
Mean	0.349	75th percentile	0.34
Standard deviation	0.807	90th percentile	0.83
Minimum	0.00	99th percentile	3.54
25th percentile	0.03	Maximum	20.32
Median (50th)	0.12	Games ≥ 1 million	1,505

4.7.5 Visualization

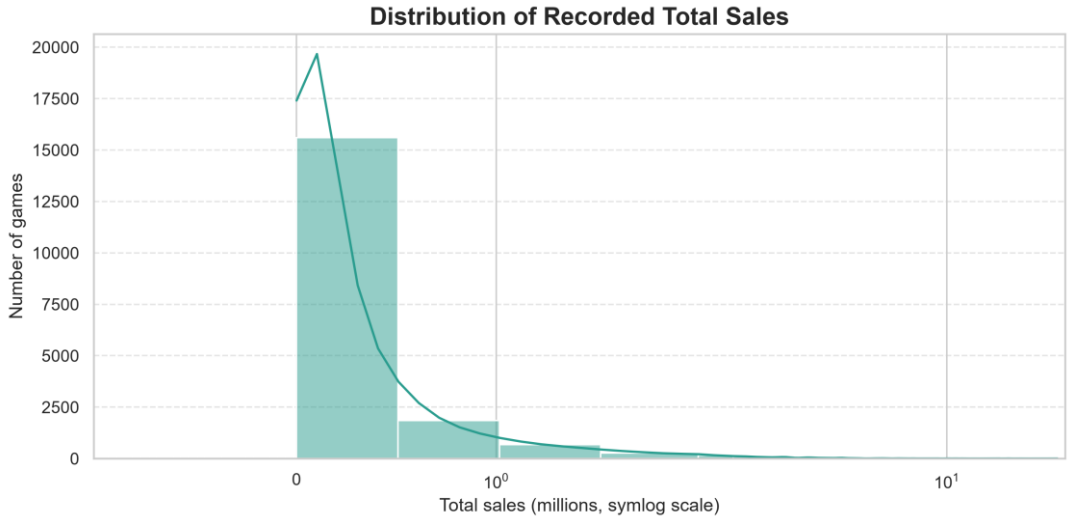


Figure 18: Distribution of recorded total sales

4.8 Analysis 8: Correlation Between Regional Sales

4.8.1 Introduction

If a game that sells well in one region also tends to sell well elsewhere, regional sales will be positively correlated. A correlation analysis quantifies how similar the four regional markets are in their taste.

4.8.2 General Description

The Pearson correlation coefficient is computed for every pair of regional sales columns using pandas' pairwise-complete handling of missing values and displayed as an annotated heat map with values between 0 and 1.

4.8.3 Specific Requirements, Functions and Formulas

- **Columns required:** na_sales, jp_sales, pal_sales, other_sales.
- **Functions:** *DataFrame.corr()* and *sns.heatmap(annot=True, fmt=".2f", cmap="YlGnBu")*.

$$r_{xy} = cov(X, Y) / (\sigma_X \sigma_Y), \quad -1 \leq r_{xy} \leq 1$$

```
sns.heatmap(df[region_columns].corr(), annot=True, cmap="YlGnBu",
            fmt=".2f", linewidths=0.5, square=True)
```

4.8.4 Analysis Results

Three of the four markets are strongly related: PAL and Other sales have the highest correlation (0.82), and North America correlates at 0.68 with PAL and 0.69 with Other. Japan is the clear exception, with correlations of only 0.07 (North America), 0.13 (PAL) and 0.08 (Other). Japanese buyers therefore appear to favour a different set of titles than the other regions, whose sales are driven largely by the same global franchises; this is consistent with a distinct local market preference. For modelling, this suggests that a model trained on global totals will mainly learn Western market behaviour.

Table 15: Pearson correlation matrix of regional sales

	na_sales	jp_sales	pal_sales	other_sales
na_sales	1.00	0.07	0.68	0.69
jp_sales	0.07	1.00	0.13	0.08
pal_sales	0.68	0.13	1.00	0.82
other_sales	0.69	0.08	0.82	1.00

4.8.5 Visualization

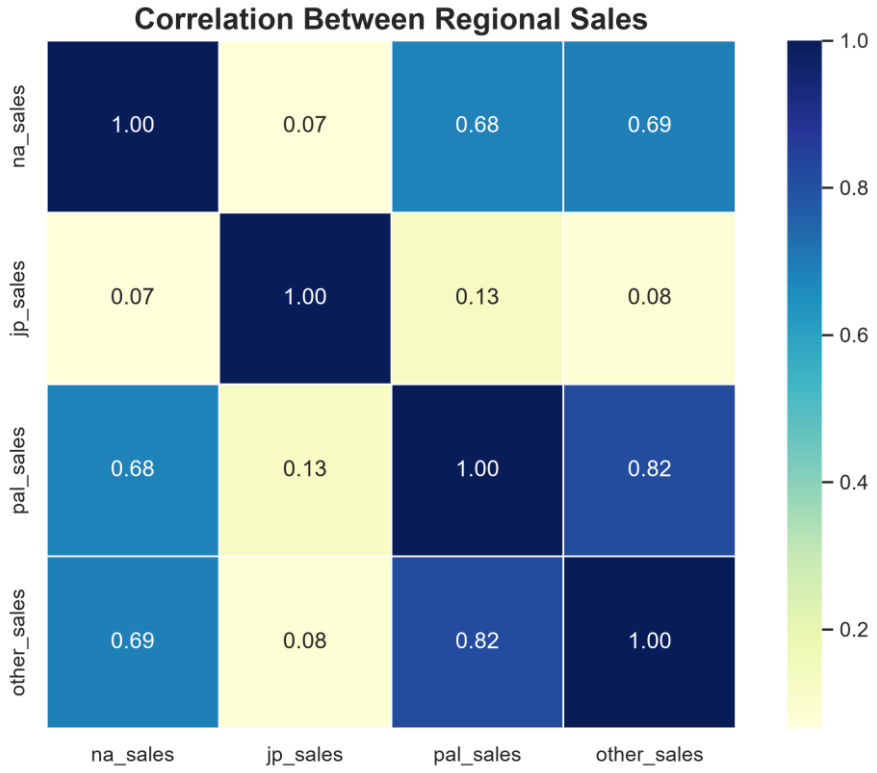


Figure 19: Correlation between regional sales

4.9 Analysis 9: Predicting Total Sales with Machine Learning

4.9.1 Introduction

The final analysis moves from description to prediction. The goal is to estimate a game's total sales from information that is available around launch, and to find out which of four common regression algorithms performs best.

4.9.2 General Description

The **target** variable is *total_sales*. The **features** are *critic_score* and *release_year* (numeric) and *console* and *genre* (categorical). Only the 18,922 games with a recorded total sales value are used. Of these, 4,126 (21.8%) have a critic score and 18,821 (99.5%) have a release year, so missing critic scores are imputed and the model effectively relies on platform, genre and year for most records. The data is split randomly into 80% training (15,137 games) and 20% test (3,785 games) with *random_state = 42*. All preprocessing is placed inside a scikit-learn *Pipeline*, so that imputation values, scaling parameters and encoding categories are learnt from the training data only, which prevents data leakage into the test set.

4.9.3 Specific Requirements, Functions and Formulas

Preprocessing. Numeric features are imputed with the median and standardised; categorical features are imputed with the most frequent value and one-hot encoded (unseen categories are ignored at prediction time).

$$\text{Standardisation: } z = (x - \mu) / \sigma$$

```
numeric_preprocessor = Pipeline([
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler())])
categorical_preprocessor = Pipeline([
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("onehot", OneHotEncoder(handle_unknown="ignore"))])
preprocessor = ColumnTransformer([
    ("numeric", numeric_preprocessor, numeric_features),
    ("categorical", categorical_preprocessor, categorical_features)])
```

Algorithms. Table 16 lists the four regressors and their settings. KNN predicts the average of the *k* most similar training games; Linear Regression fits a weighted sum of the features by least squares; the Decision Tree recursively splits the data to minimise squared error; and the Random Forest averages many such trees built on bootstrap samples.

Table 16: Regression models and hyper-parameters

Model	Principle	Settings
KNN	Average of the nearest neighbours in feature space	n_neighbors = 5 (Euclidean distance)
Linear Regression	Ordinary least squares on the encoded features	Default parameters
Decision Tree	Recursive binary splits minimising squared error	max_depth = 12, random_state = 42
Random Forest	Ensemble of bootstrap-trained trees, averaged	n_estimators = 100, max_depth = 15, random_state = 42

$$\text{KNN prediction: } \hat{y} = (1/k) \sum_{j \in N_k(x)} y_j$$

$$\text{Linear regression: } \hat{y} = \beta_0 + \beta_1 x_1 + \ldots + \beta_p x_p$$

Evaluation metrics. The models are compared with three standard regression metrics computed on the test set of n games:

$$MAE = (1/n) \sum |y_i - \hat{y}_i|$$

$$RMSE = \sqrt{[\ (1/n) \sum (y_i - \hat{y}_i)^2\ ]}$$

$$R^2 = 1 - \sum (y_i - \hat{y}_i)^2 / \sum (y_i - \bar{y})^2$$

MAE is the average absolute error in millions of units. RMSE penalises large errors more heavily and is the criterion used to choose the best model. R^2 measures the share of variance explained; a value of 0 equals predicting the mean for every game and a negative value is worse than that baseline.

4.9.4 Analysis Results

Table 17 and Figure 20 show the performance of the four models on the test set. **KNN achieves the lowest RMSE (0.760 million) and the highest R^2 (0.210)**, followed closely by Random Forest (RMSE 0.769, R^2 0.191). Random Forest has the lowest MAE (0.304), meaning it is slightly more accurate for typical games, whereas KNN makes slightly smaller errors on the few large hits, which RMSE penalises most. Linear Regression is clearly weaker (RMSE 0.807, R^2 0.109) because the relationship between the features and sales is not linear. The Decision Tree (depth 12) overfits the training data and performs worst, with an RMSE of 0.965 and a **negative R^2 of −0.273**, i.e. it is worse than always predicting the mean sales.

Table 17: Model performance on the test set (sorted by RMSE)

Model	MAE	RMSE	R^2
KNN	**0.310**	**0.760**	**0.210**
Random Forest	0.304	0.769	0.191
Linear Regression	0.363	0.807	0.109
Decision Tree	0.340	0.965	−0.273

Since $R^2 = 0.21$ for the best model, the implied standard deviation of test sales is about 0.855 million; KNN reduces the typical squared error by roughly 21% compared with predicting the mean, which is useful but modest. The reasons are visible in the earlier analyses: sales are extremely skewed (Analysis 7), the available features contain no information on franchise, publisher or marketing, about four fifths of the critic scores had to be imputed, and the target is available only for a non-random 30% of all games. *Note:* the values above are taken from the saved file *Model_Evaluation_Results.csv*; re-running the script with a different library version can change the third decimal place (for example, KNN RMSE of 0.761–0.763), but the ranking of the models stays the same.

4.9.5 Visualization

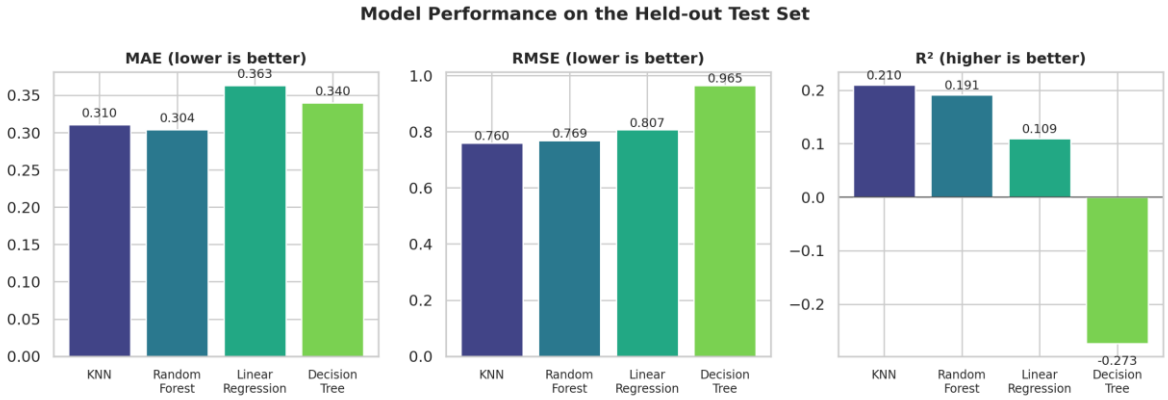


Figure 20: Comparison of MAE, RMSE and R^2 for the four models

5. Conclusion

5.1 Summary of the Work

This project analysed 64,016 video game records from 1980 to 2024 using Python. The raw data was profiled, validated and cleaned; five features were engineered; ten visualizations were produced; and four regression algorithms were compared for predicting total sales. The entire workflow is reproducible from a single script and is published, together with the data and figures, in a public GitHub repository.

5.2 Key Findings

1. **Data quality.** The dataset has no duplicates and no invalid scores or negative sales, but is highly incomplete: 70.44% of total sales and 89.57% of critic scores are missing. Conclusions therefore apply mainly to the minority of games with recorded values.
2. **Genres.** "Misc" (9,304), Action (8,557) and Adventure (6,260) are the most numerous genres, but Shooter has the highest reliable average sales (0.670 million per game), and Sports, Action and Shooter generate the largest total sales.
3. **Regions.** North America accounts for about 51% of recorded sales and PAL for 29%. Japan behaves differently from other regions, with correlations of only 0.07–0.13 with them, while PAL, Other and North America are strongly correlated (0.68–0.82).
4. **Time trend.** Releases peaked in 2009 with 4,360 games; the 2000s and 2010s hold about three quarters of the dated records. The decline after 2020 mainly reflects incomplete recent data.
5. **Critic reception.** Higher critic scores go with higher sales ($r \approx 0.28$). High-rated games average 1.00 million units, about four times the 0.24 million of low-rated games.
6. **Platforms.** Among platforms with large catalogues, Xbox 360, PS3 and PS4 have the highest average sales per game (0.59–0.66 million); the averages of GBC and NES (platforms) and of Sandbox (genre) rest on very few titles.
7. **Sales distribution.** Sales are extremely right-skewed: median 0.12 million, mean 0.349 million, maximum 20.32 million. A handful of franchises such as *Grand Theft Auto* and *Call of Duty* dominate the top of the market.

8. **Prediction.** KNN gave the best result on the test set (MAE 0.310, RMSE 0.760, R^2 0.210), followed by Random Forest (RMSE 0.769). Linear Regression and especially the Decision Tree ($R^2 = -0.273$) were weaker. Predictive power is real but modest.

5.3 Recommendations

Based on the findings of Chapter 4, the following recommendations are made for publishers, developers and analysts. They are based on associations in the recorded data and should be read together with the limitations in Section 5.4.

- **Invest in quality and reception.** Games with a High critic score (above 7) average 1.00 million units against 0.24 million for Low-rated games, so quality assurance, polish and review readiness are worth prioritising for titles with large budgets.
- **Favour proven high-volume genres.** Shooter (0.670 million per game), Action-Adventure (0.563) and Sports (0.457) combine high averages with large recorded totals. Newcomers should expect strong competition, since these genres are dominated by a few franchises.
- **Plan for North America and PAL first.** These regions account for about 51% and 29% of recorded sales, so release schedules, marketing budgets and localisation should be planned around them.
- **Treat Japan as a separate market.** Japanese sales correlate only 0.07–0.13 with other regions, so titles aimed at Japan need their own genre mix, localisation and marketing instead of a global copy of the Western plan.
- **Choose current-generation platforms.** Xbox 360, PS3, PS4 and Xbox One have average sales of about 0.51–0.66 million per title, higher than PS2 and PS (about 0.48 million), which favours platforms with an active installed base.
- **Use the prediction model only as a baseline.** With $R^2 = 0.21$, the KNN model is useful for rough screening but not for budget decisions. Franchise, publisher strength, marketing spend and price should be added and a log-transformed target and cross-validation used before relying on it (Chapter 6).
- **Improve data collection.** Sales are recorded for only 30% of titles and critic scores for about 10%. Complete records, especially for recent and digital-only releases, would make future trend analysis, including the apparent fall after 2020, more reliable.

5.4 Limitations

- **Missing target values.** Models were trained only on the 29.6% of games with recorded total sales, which is not a random sample of the market.
- **Imputation.** About 78% of the modelled games had no critic score, so the median was imputed; the feature therefore contributes little for most records.
- **Limited features.** Publisher, developer, franchise, marketing, price and user reviews were not used as predictors although they are likely to matter.
- **Evaluation design.** A single 80/20 random split and fixed hyper-parameters were used; no cross-validation, tuning or log-transformation of the skewed target was performed. Results can vary slightly between runs and library versions.
- **Small groups.** Averages for Sandbox (one game), GBC (three games) and some other categories are statistically unreliable.

5.5 Project Deliverables and Public Availability

The complete project—script, notebook, raw/cleaned/featured datasets, model results and all ten figures—is hosted in the public GitHub repository *Himanshu-56-design/Preditive_Analysis_CA2* (Figure 21), available at **https://github.com/Himanshu-56-design/Preditive_Analysis_CA2**. The repository contains 23 commits, all made within a single week as shown in the commit-activity graph (Figure 22). A summary of the work was also shared as a LinkedIn post (Figure 23; **https://lnkd.in/p/gUUVf4JF**) to communicate the results to a wider audience.

Github - https://github.com/Himanshu-56-design/Preditive_Analysis_CA2

Linkedin - **https://lnkd.in/p/gUUVf4JF**

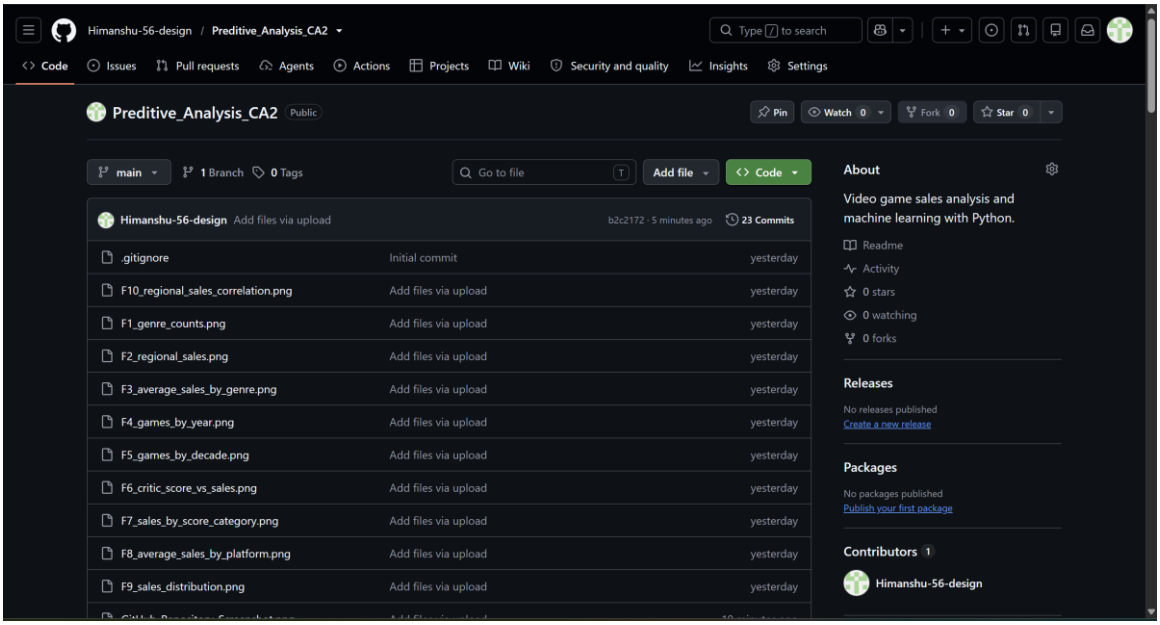


Figure 21: GitHub repository of the project (Preditive_Analysis_CA2)

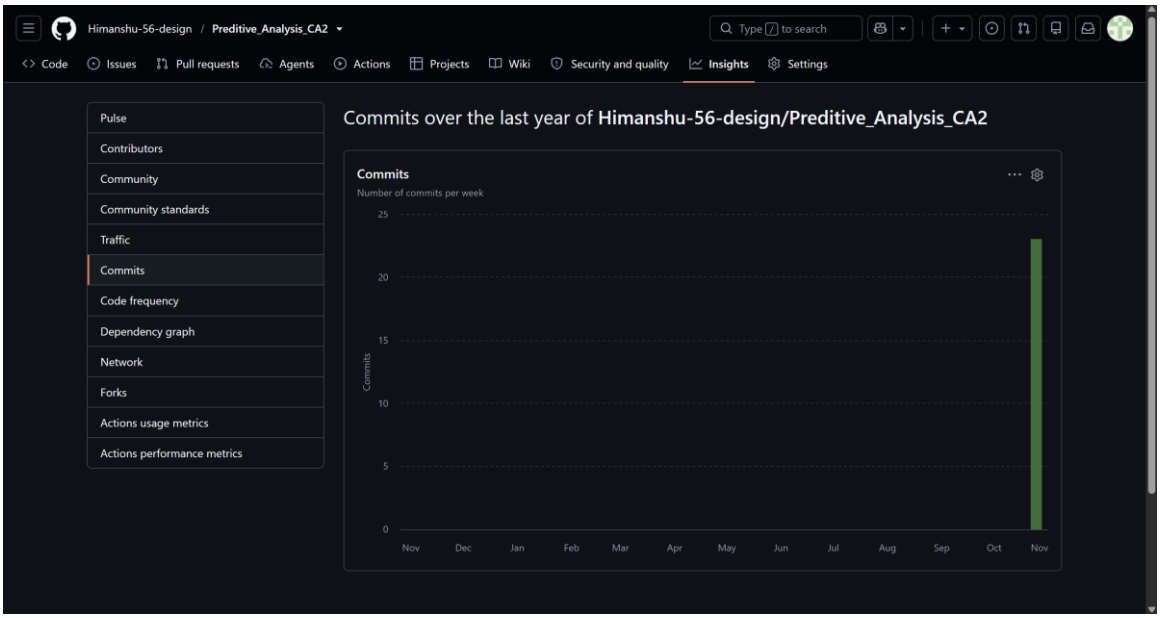


Figure 22: Commit activity of the repository (GitHub Insights)

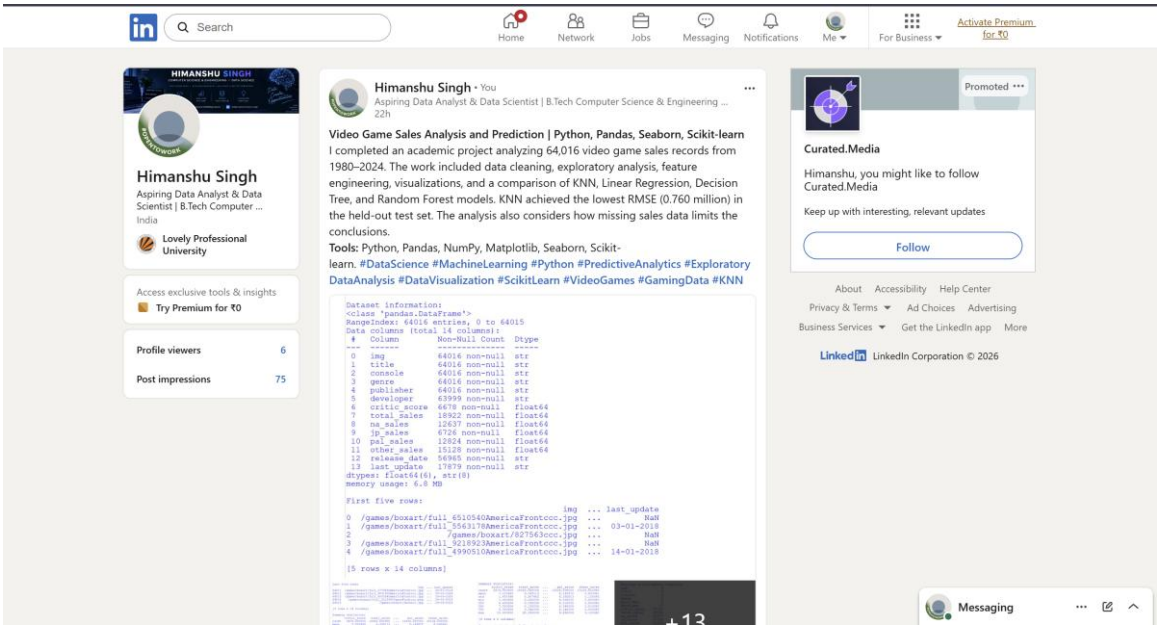


Figure 23: LinkedIn post summarising the project

5.6 Learning Outcomes

Through this project the following skills were practised: structured data profiling and documentation of cleaning decisions; feature engineering; choosing suitable chart types and scales for skewed data (symlog axes); building leak-free preprocessing pipelines; comparing models with appropriate metrics; and communicating technical results honestly, including their limitations.

6. Future Scope

The present work provides a baseline that can be extended in several directions:

1. **Better target handling.** Apply a logarithmic transformation (log(1 + sales)) to the skewed target and evaluate with metrics such as RMSLE or median absolute error so that the model is not dominated by a few blockbusters.
2. **Stronger algorithms and tuning.** Evaluate gradient boosting methods (XGBoost, LightGBM, CatBoost), and use k-fold cross-validation with *GridSearchCV* or *RandomizedSearchCV* to tune *k*, tree depth and forest size. Pruning or a smaller depth would reduce the Decision Tree's overfitting.
3. **Richer features.** Add publisher and developer statistics (for example average past sales), franchise or sequel indicators, release month/season, platform generation, user scores and text features derived from game titles using TF-IDF or embeddings.
4. **Missing-data strategies.** Use model-based imputation (KNN or iterative imputation) for critic scores, add "missing" indicator variables, and build a two-stage model that first predicts whether sales will be recorded.
5. **Classification of hits.** Reformulate the task as predicting whether a game will exceed one million units, which is more robust to the skew and easier to evaluate with precision, recall and ROC-AUC.
6. **Time-aware validation and forecasting.** Replace the random split with a chronological split (train on older games, test on newer ones) and apply time-series methods to forecast platform- or genre-level sales.
7. **Regional models.** Build separate models for North America, PAL and Japan, given the low correlation of Japan with other regions.
8. **Deployment and dashboards.** Package the best pipeline with *joblib*, expose it through a Streamlit or Flask web application and create an interactive Power BI or Plotly dashboard for non-technical users.

7. References

[1] McKinney, W. (2010). Data structures for statistical computing in Python. *Proceedings of the 9th Python in Science Conference*, 56–61.

[2] Harris, C. R., Millman, K. J., van der Walt, S. J., et al. (2020). Array programming with NumPy. *Nature*, 585, 357–362.

[3] Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3), 90–95.

[4] Waskom, M. L. (2021). seaborn: statistical data visualization. *Journal of Open Source Software*, 6(60), 3021.

[5] Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

[6] Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.

[7] Cover, T., & Hart, P. (1967). Nearest neighbor pattern classification. *IEEE Transactions on Information Theory*, 13(1), 21–27.

[8] Breiman, L., Friedman, J., Olshen, R., & Stone, C. (1984). *Classification and Regression Trees*. Wadsworth.

[9] Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning* (2nd ed.). Springer.

[10] James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning* (2nd ed.). Springer.

[11] Tukey, J. W. (1977). *Exploratory Data Analysis*. Addison-Wesley.

[12] Divekar, B. (n.d.). *Video Games Sales EDA* [Kaggle notebook and input dataset: Video Games Sales (1980-2024) - Raw.csv]. Kaggle. https://www.kaggle.com/code/bhushandivekar/video-games-sales-eda. Accessed on 28/10/2026.

[13] scikit-learn developers. *scikit-learn User Guide* and API reference. https://scikit-learn.org/stable/user_guide.html

[14] pandas development team. *pandas documentation*. https://pandas.pydata.org/docs/

[15] Singh, H. (2026). *Preditive_Analysis_CA2: Video game sales analysis and machine learning with Python* [Source code repository]. GitHub. **https://github.com/Himanshu-56-design/Preditive_Analysis_CA2**

[16] Singh, H. (2026). *Video Game Sales Analysis and Prediction | Python, Pandas, Seaborn, Scikit-learn* [LinkedIn post]. LinkedIn. **https://lnkd.in/p/gUUVf4JF**